Street-based neighborhood perception

Transportation and Urban Studio E

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Context

This case study examines which streets people regard as belonging to a named neighborhood. It links individual cognitive maps to street location, geographic boundaries and visual streetscape features. The outcome is a binary inclusion judgment for each person-street pair; people can include many streets in their perceived neighborhood. [1, 2]

Data Collection

On-street surveys were conducted on 20, 21 and 23 October 2020 in Kiyosumi-Shirakawa and Kiba, Tokyo. Respondents drew the area they associated with each neighborhood name on a map. The original study collected 25 residents and 25 visitors in each neighborhood, or 100 respondents in total. Reference points were near the corresponding metro stations and major parks. [1]

The full research dataset crosses 50 respondents with 258 streets in Kiyosumi-Shirakawa and 473 streets in Kiba, yielding 36,550 person-street observations. The local class sample **street_perception_sample.csv** contains **5,000 rows and 42 columns**, covering all 100 respondents and all 731 city-street combinations.

Choice outcome and identifiers

Field / key	Meaning
`included`	1 = the respondent includes the street in the perceived neighborhood; 0 = does not include it.
`(city, resident, person)`	Respondent key. person = 1-25 is reused across cities and resident/visitor groups. This composite key identifies 100 people.
`(city, street)`	Street key. Street numbers are reused across the two study areas.
`(city, resident, person, street)`	Observation key: unique for all 5,000 local rows.

A row is a binary judgment, not one member of a mutually exclusive set of candidate streets. Summing included within a respondent does not generally give one. A binary logit is a natural starting point; panel and spatial models can account for dependence across judgments. [2]

Use conditions

Classroom use only. The GroupProject teaching material identifies this dataset as confidential: use it only for this class and remove it from your computer after the group project.

Variables: street attributes and outcome

Name	Description / coding
city	Kiyosumi (Kiyosumi-Shirakawa) or Kiba.
street / person	Street ID / respondent number within city and resident/visitor group. Use the composite keys on page 1.
included	Binary neighborhood-inclusion response (1 yes, 0 no).
distance_m / distance_km	Network shortest-path distance from the neighborhood reference point to the nearer endpoint of the street, in metres / kilometres. These are two units of the same measure. [1]
tree / building / road / sky / other	Percentages of image pixels classified as trees, buildings, road, sky and other features. Values use a 0-100 scale and sum to 100 in every local row.
along_park	Indicator for a street along a park.
inner_river_0	Indicator for a street inside the reference river boundary.
inner_area	Indicator for a street inside the address-based neighborhood boundary.
inner_avenue_0	Indicator for a street inside the reference avenue boundary.
cos_simil	Cosine similarity of the street's visual composition to the neighborhood reference composition. Larger values mean greater similarity.
euclid_dist	Euclidean distance between visual-composition vectors. Larger values mean greater visual dissimilarity.

Variables: respondent characteristics

Name	Description / coding
resident / visitor	Indicators for the two survey groups. Residents lived within the survey basemap area; visitors did not. These sum to 1.
male / female	Gender indicators. These sum to 1 in the local sample.
age_0_20 / age_30 / age_40 / age_50 / age_60+	Age-band indicators: below 30, thirties, forties, fifties, and 60 or older.
Lower_40 / Upper_40	Age below 40 / age 40 or older. These duplicate aggregated information from the age-band fields.
under_year / year_1_5 / year_5_10 / year_10_20 / year_20+	Residence-duration bands: under one year; 1-5; 5-10; 10-20; 20+ years. Applicable to residents; all are zero for visitors. Exact boundary inclusivity is not given in the class codebook.
Under_10_years / Upper_10_years	Aggregated residence-duration indicators. Under_10_years combines the first three duration bands; Upper_10_years combines the last two. Both are zero for visitors.

Variables: familiarity and supporting files

Name	Description / coding
`everyday / 2_3_days_per_week / 1_day_per_week / 2_3_days_per_month / 1_month`	Visitor-frequency indicators: every day; 2-3 days/week; 1 day/week; 2-3 days/month; approximately monthly. Applicable to visitors; all are zero for residents. The exact endpoint of the lowest-frequency category is not documented in the class codebook.
`Under_1_day / Upper_1_day`	Aggregated visit-frequency indicators: less than weekly / at least weekly. Under_1_day combines the two monthly groups; Upper_1_day combines the three weekly/daily groups. Both are zero for residents.

Auxiliary street-level CSV and GIS data

The folder **GIS_for_street_data** contains street_data_kiyosumi.csv (258 rows) and street_data_kiba.csv (473 rows), each with 43 columns. These are one-row-per-street tables, not additional person-level choices. Link them to the main sample using (city, street). Shapefiles are supplied under gis/kiyosumi_shp and gis/kiba_shp; geopandas_vis.py gives an example of visualization, with paths that may require adjustment.

Auxiliary fields	Meaning / use
`street, city; distance and visual variables`	Street identifiers and attributes overlapping the main sample. Use the columns from one source consistently when merging.
`included / not_included / incl_prob`	Counts of inclusion / non-inclusion among the original 50 respondents in that city, and the resulting inclusion share. These are aggregated outcomes, not street characteristics known independently of responses.
`inner_river_1 to _3; inner_avenue_1 / _2`	Additional boundary indicators. Their precise spatial definitions require the original GIS layers or study documentation.
`cos_simil100/250/500; euclid_dist100/250/500`	Additional similarity/distance variants. The suffix definitions are not specified in the class codebook; do not assume units or reference areas from the names alone.
`total / extra; cluster / cluster_type`	Additional image-processing fields and streetscape cluster labels. Definitions and cluster-label mapping require the source processing documentation.
`betweenness_centrality`	Stored street-network centrality measure; check its normalization before comparing with independently computed centrality.
`u_osmid / v_osmid; u_lat / u_lon / v_lat / v_lon`	Identifiers and geographic coordinates of street endpoints.
`u_angle / v_angle / ave_angle`	Stored endpoint/average angle measures. Confirm the reference direction and units before directional modeling.

Do not substitute the auxiliary included count for the main binary included outcome. Using incl_prob as a predictor can leak responses from the original survey into the model, including evaluation data.

Descriptive statistics

Choice shares use person-street rows as their denominator, not respondents or unique streets.

Study area / outcome	Count	Share within area
Kiyosumi: included	520	29.45%
Kiyosumi: not included	1,246	70.55%
Kiyosumi total	1,766	100.00%
Kiba: included	832	25.73%
Kiba: not included	2,402	74.27%
Kiba total	3,234	100.00%
Combined: included	1,352	27.04%
Combined: not included	3,648	72.96%
Combined total	5,000	100.00%

Variable	N	Min	Max	Mean	SD
`distance_km`	5,000	0	1.963	0.86	0.38
`tree`	5,000	0	52	7.69	9.05
`building`	5,000	1	72	41.95	13.82
`road`	5,000	3	55	36.72	6.74
`sky`	5,000	0	46	10.87	5.92
`cos_simil`	5,000	0.402298	0.997212	0.82	0.10
`euclid_dist`	5,000	3.98434	64.7765	34.16	12.05

SD is the sample standard deviation. Street attributes are repeated for multiple people, so these are row-weighted statistics of the sampled judgments. For a street-level summary, deduplicate (city, street) or use the auxiliary tables. The sample contains 2,450 resident rows and 2,550 visitor rows, with 25-77 sampled streets per respondent.

Preparation notes

Use a composite respondent ID before fitting panel models or splitting the data by person. Retain city in all street joins. Age, gender and familiarity indicators are redundant within their groups; omit a reference category when fitting an intercept. Residence-duration zeros for visitors and visit-frequency zeros for residents are structural, not evidence of zero familiarity.

For spatial models, construct adjacency or distance weights from the relevant city network and align the matrix with street IDs explicitly. The sampled rows do not form a complete person-by-street grid; preserve the observed-row mask rather than filling absent judgments with zero.

Research applications and sources

A binary logit can relate neighborhood inclusion to distance, boundaries and streetscape composition. Mixed or hierarchical models can represent differences between respondents and study areas. Spatial dependence among neighboring streets can be incorporated using the methods in [2]. Because visual percentages sum to 100, including all five components together with an intercept creates exact collinearity.

References and sources

[1] Oyama, Y. (2024). Spatial city image and its formative factors: A street-based neighborhood cognition analysis. Cities, 149, 104898. Local copy: street_perception_paper1.pdf.
https://doi.org/10.1016/j.cities.2024.104898

[2] Oyama, Y., Murakami, D. and Krueger, R. (2024). A hierarchical Bayesian logit model for spatial multivariate choice data. Journal of Choice Modelling, 52, 100503. Local copy: street_perception_paper2.pdf.
https://doi.org/10.1016/j.jocm.2024.100503

Source pages checked on 10 November 2026. Counts and descriptive statistics refer to the local CSV files, excluding headers. This teaching description was prepared from the cited sources and local file inspection; it is not an author-supplied codebook.

Local teaching sources: GroupProject.pdf / GroupProject.pptx, street_perception_description.txt, street_perception_sample.csv and the supporting GIS_for_street_data folder.